Mir Shafayeth Jamil

shafayeth.cu.physics@gmail.com | Chattogram, Bangladesh | ORCID: 0009-0006-4271-840X

RESEARCH INTERESTS

Horizon thermodynamics and the thermodynamic derivation of the gravitational field equations; emergent and modified gravity. The Raychaudhuri equation, geodesic focusing, energy conditions, and the singularity theorems. Gravity with a preferred frame and the initial value formulation of modified gravity: hyperbolicity, constraint propagation, and well-posedness. Quantum gravity, including discrete approaches to spacetime structure such as causal set theory; the foundations of quantum theory.

EDUCATION

Master of Science (M.S.) in Applied Mathematics and Theoretical Physics *2024 – 2025*
Jamal Nazrul Islam Centre for Advanced Research (JNICAR)
University of Chittagong, Chattogram, Bangladesh **CGPA: 3.67 / 4.00 (Ranked 2nd of 16)**

Bachelor of Science (B.Sc.) in Physics *2017 – 2023*
University of Chittagong, Chattogram, Bangladesh **CGPA: 2.95 / 4.00**

IELTS Academic (December 2025) **Overall Band 8.0**
Reading 9.0 | Listening 8.5 | Writing 7.5 | Speaking 7.0

RESEARCH EXPERIENCE

Independent Research *2025 – Present*

Charge Conservation Fixes the Expansion of Bohmian Congruences *(under review, 2026)*

- Identified the normalized de Broglie–Bohm congruence, introduced but left undeveloped in the Lashin–Dou critique, as the unit flow of the conserved Klein–Gordon current, so that its expansion is fixed algebraically by charge conservation and carries no independent initial datum; the quantum Raychaudhuri equation is therefore the once-differentiated continuity equation and supplies no dynamics beyond it.
- Showed that the congruence is a gradient flow with vanishing vorticity, so the no-crossing property invoked in support of defocusing removes the only anti-focusing term rather than supplying one.
- Established, with the field treated as a gravitating source, that the quantum potential cannot reverse null focusing for any state, since the null contraction removes its explicit term in the part of the stress tensor proportional to the metric and it survives only through a positive effective-mass factor, and that the timelike convergence is bounded below by the classical mass-term value; in minimally coupled flat FRW, charge conservation imposes a dust-like floor on the energy density that forbids every bounce and drives the scale factor to zero in finite proper time. Scope stated explicitly: exact under FRW symmetry and minimal coupling only.

Hyperbolicity and the Initial Value Formulation of an Einstein–Entropy–Clock Model *(in preparation, 2026)*

- Studied the initial value problem for a modified gravity model in which gravity couples to an entropy scalar and a clock field whose normalized gradient defines a preferred timelike flow, a hypersurface-orthogonal preferred frame with the normalization imposed by a Lagrange multiplier, with a derivative coupling between the entropy sector and the expansion of that flow. Reduced the system to spherical symmetry, wrote it in quasilinear first-order form with an eleven-component state vector, and computed the principal symbol; the coupling mixes the entropy and clock sectors at principal level rather than only through lower-order terms.
- Showed that the constraints propagate consistently: the integrability constraints as kinematic identities, the clock normalization by construction, and the Hamiltonian and momentum constraints through a closed first-order system following from the contracted Bianchi identity and Noether's second theorem, so the derivative coupling produces no anomalous divergence of the stress tensor.
- Found that strong hyperbolicity holds only pointwise, on a codimension-one semisimplicity locus in background state space; off that locus the characteristic matrix carries a nontrivial Jordan block and strong hyperbolicity fails. Since the locus does not contain an open neighbourhood, the standard quasilinear existence theorems do not apply, and the local well-posedness statement remained conditional. Left the manuscript unsubmitted rather than overstate the result.

M.S. Thesis *JNICAR, University of Chittagong, 2024 – 2025*

Galactic Dynamics and Cosmological Observables without Particle Dark Matter: A Comparative Study of MOND (AQUAL/QUMOND/TeVeS) and Emergent Gravity

Supervisor: Prof. Anjan Kumar Chowdhury

- Worked through the thermodynamic route to gravitational dynamics in full: the Gibbs state obtained by maximising von Neumann entropy, the KMS condition, the Unruh effect, local Rindler thermodynamics, and Jacobson's derivation of the Einstein field equations from the Clausius relation, including the local equilibrium conditions the derivation requires and its silence on the cosmological constant.
- Examined Verlinde's entropic force argument and his de Sitter emergent gravity, in particular the volume contribution departing from the area law, together with its galactic and cosmological consequences.
- Compared the MOND family and emergent gravity against galactic and cosmological observables, and identified the absence of a covariant action principle as the structural limitation of the emergent framework.

PAPERS AND PREPRINTS

- **M. S. Jamil**, "Charge Conservation Fixes the Expansion of Bohmian Congruences," under review at the *Canadian Journal of Physics* (submitted September 2026). Manuscript available on request.
- **M. S. Jamil**, "Hyperbolicity and the Initial Value Formulation of an Einstein–Entropy–Clock Model," in preparation, 2026. Draft available on request.
- **M. S. Jamil**, "Galactic Dynamics Without Dark Matter: A Phenomenological Comparison of MOND and Emergent Gravity," preprint, *Preprints.org*, December 2025. DOI: 10.20944/preprints202512.1605.v1.

SELECTED M.S. COURSEWORK

Advanced Quantum Mechanics | Classical Theory of Fields | Quantum Field Theory | General Relativity and Gravitation

ACADEMIC SCHOOLS AND ADVANCED TRAINING

- **ICTP PWF Physics for Bangladesh School on General Relativity** (2025): tensor calculus on manifolds, geodesics, the Riemann tensor, Einstein's field equations, the Schwarzschild solution, and FLRW cosmology.
- **PWF Bangladesh School on Quantum Field Theory** (2024): canonical quantisation of the Klein–Gordon and Dirac fields, gauge invariance, QED, Feynman diagrammatics, and the path integral.
- **IUB DPS Summer School on Quantum Mechanics and Topological Physics of Matter** (2024): group-theoretic foundations of quantum mechanics, quantum information, and topological phases of matter.

MATHEMATICAL AND COMPUTATIONAL SKILLS

Mathematical Methods: Differential geometry, vector and tensor calculus, linear algebra, real and complex analysis, ordinary and partial differential equations, Fourier analysis and integral transforms, group theory, multivariable calculus.

Programming and Software: Python (NumPy, SciPy, Matplotlib), C, LaTeX, Mathematica (working knowledge).

TEACHING EXPERIENCE

Private Tutor, Mathematics and Physics *Chattogram, Bangladesh, 2017 – 2025*

Eight years of one-to-one and small-group instruction in calculus, vectors, trigonometry and complex numbers, and in Newtonian and rotational mechanics, waves and thermodynamics.

ACADEMIC REFEREES

Prof. Anjan Kumar Chowdhury *M.S. thesis supervisor*

Professor, Jamal Nazrul Islam Centre for Advanced Research, University of Chittagong, Chattogram 4331, Bangladesh

anjan.kumar@cu.ac.bd

Dr. Md. Saiful Alam

Associate Professor, Department of Physics, University of Chittagong, Chattogram 4331, Bangladesh

saif10350@cu.ac.bd